

Phosphate plant -- Diagnostics

Section report: Diagnostics

Site	Atlas Phosphate Complex (:atlas_phosphate)
Reporting period	2026-01-01T00:00:00 to 2026-12-31T23:00:00 (365 days, 8568 operating hours)
Product	Dap from 29.0 % P2O5 ore, 1713.0 kt of product
Framework scope	ISO 50001:2018 energy management; ISO 14001:2015 environmental management; ISO 46001:2019 water efficiency; ISO 9001:2015 quality management; GHG Protocol corporate standard; EU Industrial Emissions Directive 2010/75; US EPA 40 CFR 63 subpart AA; IFA/EFMA BAT for phosphate fertilisers; Phosphogypsum stack management code
Provenance	Design revision A seed 20260101 8760 hourly intervals of 69 signals generated 2026-09-23T15:43:11.110 basis :screening_plant_model

Fault detection and diagnostics

DIAGNOSTIC SUMMARY

Quantity	Value
Findings	100 (30 critical, 65 warning, 5 info)
Value at stake	1.67734421e8 :USD a year
Hours in alarm	19,447 h
Rules applied	20 rules of the catalogue
Deviations caught	8 of 15 (53.3 %)
Status	:action_required

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Critical	:flotation	:concentrate_grade_low	:CONC_P2O5	27.63	29.5	:wt_pct	6.34	2026-12-06T10:00:00	2026-12-11T03:00:00	114	1.76516e7
Critical	:flotation	:concentrate_grade_low	:CONC_P2O5	28.6	29.5	:wt_pct	3.07	2026-03-13T10:00:00	2026-03-21T00:00:00	183	8.527489e6
Critical	:evaporation	:acid_strength_low	:STRONG_ACID_P2O5	48.64	50.5	:wt_pct	3.68	2026-12-06T12:00:00	2026-12-11T03:00:00	112	3.815317e6
Critical	:evaporation	:acid_strength_low	:STRONG_ACID_P2O5	48.74	50.5	:wt_pct	3.49	2026-03-13T12:00:00	2026-03-16T03:00:00	64	3.619393e6
Critical	:attack	:sulfate_deficit	:FREE_SO4	0.7723	1.2	:wt_pct	35.64	2026-12-06T11:00:00	2026-12-11T00:00:00	110	3.226419e6
Critical	:attack	:sulfate_deficit	:FREE_SO4	0.7919	1.2	:wt_pct	34.01	2026-03-13T11:00:00	2026-03-16T01:00:00	63	3.079024e6
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.3	5	:count	5.98	2026-06-28T07:00:00	2026-06-28T16:00:00	10	574,052
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.24	5	:count	4.76	2026-06-18T01:00:00	2026-06-18T15:00:00	15	456,933
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.23	5	:count	4.68	2026-06-22T15:00:00	2026-06-23T01:00:00	11	449,616
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.23	5	:count	4.6	2026-06-30T21:00:00	2026-07-01T23:00:00	27	441,632
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.23	5	:count	4.56	2026-06-21T14:00:00	2026-06-22T09:00:00	20	437,807
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.23	5	:count	4.53	2026-06-24T13:00:00	2026-06-24T19:00:00	7	435,166
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.21	5	:count	4.22	2026-06-28T22:00:00	2026-06-29T13:00:00	16	405,194
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.2	5	:count	3.94	2026-06-19T10:00:00	2026-06-19T17:00:00	8	377,932
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.19	5	:count	3.8	2026-06-16T05:00:00	2026-06-17T07:00:00	27	364,612
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.18	5	:count	3.52	2026-06-23T12:00:00	2026-06-23T19:00:00	8	338,395
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.18	5	:count	3.51	2026-06-17T13:00:00	2026-06-17T18:00:00	6	337,031
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.17	5	:count	3.44	2026-06-26T18:00:00	2026-06-27T00:00:00	7	330,181
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.16	5	:count	3.11	2026-06-19T21:00:00	2026-06-21T10:00:00	38	298,491
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.13	5	:count	2.54	2026-06-24T21:00:00	2026-06-26T05:00:00	33	243,618
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.13	5	:count	2.51	2026-06-30T09:00:00	2026-06-30T12:00:00	4	240,763
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.12	5	:count	2.31	2026-06-29T16:00:00	2026-06-30T05:00:00	14	222,178
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.12	5	:count	2.3	2026-06-27T14:00:00	2026-06-27T16:00:00	3	221,260
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.09	5	:count	1.86	2026-06-26T11:00:00	2026-06-26T16:00:00	6	178,928
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.34	17.4	:wt_pct	6.09	2026-12-06T08:00:00	2026-12-11T06:00:00	119	145,116
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.07	5	:count	1.49	2026-06-23T21:00:00	2026-06-24T02:00:00	6	142,967
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.39	17.4	:wt_pct	5.83	2026-03-13T08:00:00	2026-03-16T05:00:00	70	139,058
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.05	5	:count	1.07	2026-06-19T01:00:00	2026-06-19T03:00:00	3	103,086
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.96	17.4	:wt_pct	2.51	2026-08-09T07:00:00	2026-08-20T03:00:00	261	59,921
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.02	5	:count	0.3091	2026-06-24T05:00:00	2026-06-24T07:00:00	3	29,673
Warning	:filtration	:filter_rate_low	:FILTER_RATE	3.32	4	t_per_m2_d	16.92	2026-12-06T16:00:00	2026-12-11T00:00:00	105	4.1379752e7
Warning	:filtration	:filter_rate_low	:FILTER_RATE	3.35	4	t_per_m2_d	16.27	2026-03-13T14:00:00	2026-03-16T00:00:00	59	3.9785148e7
Warning	:beneficiation	:grind_coarse	:CYCLONE_P80	190.9	185	:um	3.19	2026-03-16T22:00:00	2026-03-21T02:00:00	101	8.344416e6
Warning	:granulation	:scrubber_ph_low	:SCRUBBER_PH	4.77	5.6	:count	14.74	2026-12-06T06:00:00	2026-12-11T02:00:00	117	1.189036e6
Warning	:granulation	:scrubber_ph_low	:SCRUBBER_PH	4.8	5.6	:count	14.2	2026-03-13T04:00:00	2026-03-16T02:00:00	71	1.144873e6

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,428.17	4,600	:kw	18	2026-11-30T02:00:00	2026-12-06T11:00:00	154	563,154
Warning	:beneficiation	:slurry_density_high	:SLURRY_DENSITY	44.4	44	:wt_pct	0.9005	2026-10-12T01:00:00	2026-10-16T00:00:00	96	560,460
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,421.56	4,600	:kw	17.86	2026-12-28T01:00:00	2026-12-31T23:00:00	95	558,661
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,415.4	4,600	:kw	17.73	2026-04-06T01:00:00	2026-04-12T20:00:00	164	554,475
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,414.21	4,600	:kw	17.7	2026-01-01T05:00:00	2026-01-04T21:00:00	89	553,661
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,412.27	4,600	:kw	17.66	2026-10-12T02:00:00	2026-10-18T21:00:00	164	552,347
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,411.47	4,600	:kw	17.64	2026-12-14T02:00:00	2026-12-20T21:00:00	164	551,798
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,410.44	4,600	:kw	17.62	2026-11-23T02:00:00	2026-11-29T20:00:00	163	551,101
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,409.68	4,600	:kw	17.6	2026-02-02T01:00:00	2026-02-08T21:00:00	165	550,584
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,407.27	4,600	:kw	17.55	2026-01-12T01:00:00	2026-01-18T21:00:00	165	548,943
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,405.7	4,600	:kw	17.52	2026-02-23T02:00:00	2026-03-01T21:00:00	164	547,873
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,404.54	4,600	:kw	17.49	2026-04-13T02:00:00	2026-04-19T20:00:00	163	547,089
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,403.94	4,600	:kw	17.48	2026-01-05T02:00:00	2026-01-11T21:00:00	164	546,677
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,402.83	4,600	:kw	17.45	2026-11-02T02:00:00	2026-11-08T20:00:00	163	545,925
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,402.37	4,600	:kw	17.44	2026-11-16T01:00:00	2026-11-22T20:00:00	164	545,609
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,401.91	4,600	:kw	17.43	2026-04-27T01:00:00	2026-05-03T20:00:00	164	545,300
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,401.8	4,600	:kw	17.43	2026-09-07T02:00:00	2026-09-13T21:00:00	164	545,225
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,401.78	4,600	:kw	17.43	2026-03-30T01:00:00	2026-04-05T21:00:00	165	545,210
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,401.17	4,600	:kw	17.42	2026-11-09T02:00:00	2026-11-15T21:00:00	164	544,793
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,400.73	4,600	:kw	17.41	2026-05-18T01:00:00	2026-05-24T20:00:00	164	544,496
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,400.47	4,600	:kw	17.4	2026-09-14T01:00:00	2026-09-20T21:00:00	165	544,322
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,400.26	4,600	:kw	17.4	2026-01-26T02:00:00	2026-02-01T21:00:00	164	544,176
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,400.14	4,600	:kw	17.39	2026-05-04T02:00:00	2026-05-10T22:00:00	165	544,096
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,399.97	4,600	:kw	17.39	2026-02-16T02:00:00	2026-02-22T20:00:00	163	543,981
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,398.98	4,600	:kw	17.37	2026-03-23T01:00:00	2026-03-29T21:00:00	165	543,306
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,398.8	4,600	:kw	17.37	2026-10-05T01:00:00	2026-10-11T21:00:00	165	543,187
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,398.46	4,600	:kw	17.36	2026-05-25T01:00:00	2026-05-31T21:00:00	165	542,955
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,397.73	4,600	:kw	17.34	2026-04-20T02:00:00	2026-04-26T21:00:00	164	542,458
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,397.56	4,600	:kw	17.34	2026-02-09T01:00:00	2026-02-15T20:00:00	164	542,344
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,397.44	4,600	:kw	17.34	2026-12-21T01:00:00	2026-12-27T21:00:00	165	542,257
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,397.44	4,600	:kw	17.34	2026-09-28T01:00:00	2026-10-04T21:00:00	165	542,257
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,395.55	4,600	:kw	17.29	2026-09-21T01:00:00	2026-09-27T21:00:00	165	540,977
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,393.98	4,600	:kw	17.26	2026-03-09T01:00:00	2026-03-13T11:00:00	107	539,910
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,390.52	4,600	:kw	17.19	2026-10-19T01:00:00	2026-10-25T21:00:00	165	537,552
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,389.85	4,600	:kw	17.17	2026-03-02T01:00:00	2026-03-08T21:00:00	165	537,097
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,389.3	4,600	:kw	17.16	2026-10-26T01:00:00	2026-11-01T21:00:00	165	536,724
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,389.12	4,600	:kw	17.15	2026-01-19T01:00:00	2026-01-25T21:00:00	165	536,601
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,385.71	4,600	:kw	17.08	2026-05-11T00:00:00	2026-05-17T20:00:00	165	534,280
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,381.92	4,600	:kw	17	2026-08-31T06:00:00	2026-09-06T20:00:00	159	531,706
Warning	:beneficiation	:slurry_density_high	:SLURRY_DENSITY	44.36	44	:wt_pct	0.8293	2026-10-06T11:00:00	2026-10-11T23:00:00	133	516,191
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,336.31	4,600	:kw	16.01	2026-07-06T03:00:00	2026-07-19T16:00:00	326	500,689
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,277.65	4,600	:kw	14.73	2026-12-11T13:00:00	2026-12-13T21:00:00	57	460,803
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,244.57	4,600	:kw	14.01	2026-06-29T06:00:00	2026-07-05T19:00:00	158	438,308
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,244.52	4,600	:kw	14.01	2026-06-08T06:00:00	2026-06-14T16:00:00	155	438,276
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,227.56	4,600	:kw	13.64	2026-08-10T06:00:00	2026-08-16T16:00:00	155	426,738

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,227.04	4,600	:kw	13.63	2026-06-15T06:00:00	2026-06-21T16:00:00	155	426,389
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,227.04	4,600	:kw	13.63	2026-08-17T06:00:00	2026-08-23T16:00:00	155	426,388
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,225.6	4,600	:kw	13.6	2026-08-03T06:00:00	2026-08-09T15:00:00	154	425,410
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,223.12	4,600	:kw	13.55	2026-07-20T06:00:00	2026-07-26T15:00:00	154	423,719
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,220.2	4,600	:kw	13.48	2026-08-24T06:00:00	2026-08-30T16:00:00	155	421,739
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,219.38	4,600	:kw	13.46	2026-07-27T06:00:00	2026-08-02T16:00:00	155	421,179
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,217.93	4,600	:kw	13.43	2026-06-22T06:00:00	2026-06-28T15:00:00	154	420,192
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,209.92	4,600	:kw	13.26	2026-06-01T01:00:00	2026-06-07T16:00:00	160	414,748
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	4,898.23	4,600	:kw	6.48	2026-03-16T19:00:00	2026-03-22T21:00:00	147	202,798
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.28	19.2	:wt_pct	0.4204	2026-10-30T18:00:00	2026-11-01T08:00:00	39	161,276
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.28	19.2	:wt_pct	0.4041	2026-10-26T14:00:00	2026-10-29T03:00:00	62	155,009
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.26	19.2	:wt_pct	0.3373	2026-10-25T10:00:00	2026-10-26T01:00:00	16	129,379
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.26	19.2	:wt_pct	0.3141	2026-10-29T11:00:00	2026-10-30T15:00:00	29	120,479
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.25	19.2	:wt_pct	0.2402	2026-11-01T16:00:00	2026-11-01T23:00:00	8	92,140
Warning	:utilities	:pump_vibration_high	:PUMP_VIBRATION	6.82	6	:count	13.59	2026-07-21T04:00:00	2026-07-31T00:00:00	237	1,019.23
Info	:filtration	:vacuum_low	:FILTER_VACUUM	163.88	200	:mmwc	18.06	2026-12-06T06:00:00	2026-12-11T01:00:00	116	173,375
Info	:filtration	:vacuum_low	:FILTER_VACUUM	165.18	200	:mmwc	17.41	2026-03-13T06:00:00	2026-03-16T01:00:00	68	167,126
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	547.66	336	:l_ph	63	2026-01-01T05:00:00	2026-03-13T08:00:00	1,708	122,766
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	545.85	336	:l_ph	62.46	2026-12-11T13:00:00	2026-12-31T23:00:00	491	121,714
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	537.6	336	:l_ph	60	2026-03-16T13:00:00	2026-12-06T07:00:00	6,355	116,926

DID THE DETECTOR CATCH WHAT THE CAMPAIGN INJECTED?

Injected deviation	Role	Kind	Magnitude	Unit	Hours	Detected	Rule that caught it	Findings
:cooling_fouling	:reactor_temperature	:drift	6.2	:deg_c	264	0	:none	0
:sulfate_deficit	:sulfate_ratio	:bias	-1.7	:wt_pct	288	0	:none	0
:grind_coarse	:grind_p80	:bias	42	:um	192	1	:grind_coarse	1
:reagent_starvation	:concentrate_grade	:bias	-1.5	:wt_pct	264	1	:reagent_demand_high	1
:filter_cloth_blinding	:filtration_rate	:drift	-1.9	:t_per_m2_d	168	0	:none	0
:evaporator_fouling	:steam_flow	:drift	13.5	:t_ph	360	0	:none	0
:acid_strength_loss	:acid_strength	:drift	-1.6	:wt_pct	312	0	:none	0
:scrubber_breakthrough	:stack_fluoride	:bias	2.4	:count	384	1	:stack_fluoride_high	21
:mill_liner_wear	:mill_load	:drift	430	:kw	264	1	:mill_power_excess	2
:pump_cavitation	:pump_vibration	:bias	3.6	:count	240	1	:pump_vibration_high	1
:sparger_blockage	:product_n	:bias	-1.2	:wt_pct	264	1	:product_nitrogen_low	1
:crusher_gap_drift	:oversize_pct	:bias	6.5	:pct	264	0	:none	0
:rock_grade_drift	:rock_grade	:drift	1.3	:wt_pct	192	0	:none	0
:cyclone_roping	:slurry_density	:bias	6.4	:wt_pct	240	1	:slurry_density_high	2
:ammonia_overfeed	:product_n	:bias	1.1	:wt_pct	192	1	:product_nitrogen_high	5

The catalogue of the detector holds 20 rules, each with the signal it watches, the reference it compares against and the window it averages over; 'PHOSPHATE.RULE_CATALOGUE' prints them and the notebook shows them rule by rule.